Open Source Formal Verification

Introduction/Content

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Who?

- Professor: Yann Thoma
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Short CV

- 2001: Master's degree in computer science from EPFL
- 2005: PhD from EPFL (reconfigurable systems)
- 2005–2009: Lecturer at EIG (now HEPIA; digital systems and processor architecture)
- 2006–2008: Engineer for the Group of Applied Physics (UniGe)
- 2006–2008: Engineer at REDS (HEIG-VD)
- 2009– : Professor at REDS
  - Teaching digital systems, verification of embedded systems, concurrent programming and realtime programming
  - Various research projects (FPGA development, embedded systems, pharmacology)
- 2015–2018: Director of REDS
- 2018–2019: Visiting professor at UNSW
- 2025: Same training given at CERN

My goals here

- Give you a good overview of formal verification
- Show you how to use PSL to write assertions and coverage goals
- Show you Open Source tools work quite well for formal verification
- Let you practice with exercises and examples
  - The best way to learn is to practice!
- Develop a community of CERN engineers who can use formal verification in their work
- Have fun and enjoy the course

Who are you?

Training Material

You can get all the material on the following git repo:
`https://github.com/ythoma/formal-cern-26`

- Theory
  - Slides
  - Code examples
- Exercises
  - Statements
  - Code

Course Schedule

- 3 days of training
- 3.5 hours of theory per day
- 3.75 hours of exercises per day
- 30 minutes of breaks per day
- 1.25 hours of lunch per day [1]
- The next detailed schedule is likely to shift a bit

[1] Choosing units is always a challenge.

Schedule - day 1

Time	Description	Type
8:30	Introduction	Theory
8:45	Introduction to verification	Theory
09:45	Break	
10:00	Introduction to formal verification	Theory (and an exercise)
11:00	Basic PSL constructs	Theory
11:45	Combinational systems	Exercise
12:00	Lunch	
13:15	Combinational systems	Exercise
14:15	Temporal logic	Theory
15:00	Verification of a counter	Exercise
15:30	Break	
15:45	Verification of a timer	Exercise
16:30	Verification of a state machine	Exercise
17:30	End of the day	

Schedule - day 2

Time	Description	Type
08:30	Temporal logic	Theory (end)
09:00	Verification of a counter	Exercise
09:30	Verification of a timer	Exercise
10:15	Break	
10:30	Verification of a state machine	Exercise
12:00	Regular expressions	Theory
12:45	Lunch	
13:45	Regular expressions	Exercise
14:30	Parametric design	Theory
14:45	Axi module	Exercise
15:30	Break	
15:45	Coverage	Theory
16:00	Coverage	Exercise
16:30	Helper code and variables	Theory
17:00	Helper code and variables	Exercise
17:30	End of the day	

Schedule - day 3

Time	Description	Type
08:30	Regular expressions	End of Theory
08:45	Regular expressions	Exercise
09:30	Parametric design	Theory
09:45	Axi module	Exercise
10:30	Break	
10:45	Axi module	Exercise
11:30	Coverage	Theory
11:45	Coverage	Exercise
12:00	Helper code and variables	Theory
12:30	Lunch	
13:30	Helper code and variables	Exercise
14:15	Backends	Theory
14:30	Backends	Exercise
15:00	Continuous integration and scripts	Theory
15:15	Methodology insights	Theory
15:30	Break	
15:45	Use cases	Exercise and discussions
17:20	Feedback	
17:30	End of the day	